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Gesture controlled Virtual Mouse using Computer vision

Abstract

The Virtual Mouse & Quantum Assistant is an AI-powered desktop control system that eliminates the need for physical input devices. Using real-time hand tracking via MediaPipe and OpenCV, users control the cursor, perform clicks, scroll,adjust volume, and change brightness entirely through hand gestures captured by a webcam. Alongside gesture control, the system includes Quantum-a voice and text-driven AI assistant with 31 command categories, LangChain conversation memory,LLM integration and a chat UI.

Introduction

Traditional human-computer interaction relies heavily on physical peripherals - keyboards, mice, and touchpads - which pose challenges for users with mobility impairments and limit natural interaction. With advances in computer vision and machine learning, touchless gesture-based control is now feasible on standard consumer hardware.This project addresses these limitations by building a two-mode desktop assistant:

- Gesture Mode - the webcam acts as a virtual mouse; specific hand postures trigger cursor movement, left/right click,drag, scroll, volume, and brightness control in real time.
- Quantum Assistant Mode - a voice/text chatbot that dispatches 31 categories of system commands (application control, browser tabs, file search, music, timers, unit conversion, and more), powered by fuzzy matching and a cascading LLM backend.

Objectives

- To develop a real-time, touchless mouse control system using hand gesture recognition via webcam, eliminating the need for physical input devices.
- To implement voice and text-driven AI assistant capable of dispatching 31 categories of system commands with fuzzy matching for improved accuracy.
- To integrate a cascading LLM backend with conversation memory for intelligent, context-aware responses to unrecognized commands.
- To enable custom gesture training, allowing users to define and map personal gestures to system actions such as hotkeys, app launches, or text input.
- To provide an accessible and configurable interface through a browser-based chat UI with real-time settings for gesture sensitivity, cursor speed, and pinch threshold.

Methodology / System Design

The Virtual Mouse & Quantum Assistant follows a modular architecture separating gesture processing, assistant logic, and LLM integration for efficient and maintainable system design.

1.Gesture Recognition Module:
The webcam captures live frames processed by MediaPipe Hands to extract 21 hand landmarks per frame. The HandRecog class classifies these landmarks into gesture enums (V sign, Fist, Pinch, etc.) using finger state detection. Gestures are stabilized over 4+ consecutive frames to eliminate jitter.

Figure 1 :Gesture recognition and cursor movement

2. System Control Module:
The Controller class maps recognized gestures to system actions - cursor movement, left/right click, double click, drag, scroll, volume, and brightness control. PyAutoGUI handles cross-platform mouse actions, with Windows-specific audio/brightness libraries loaded conditionally.

Figure 2 :Gesture-to-action mapping interface

3. Quantum Assistant - Command Dispatcher:
User voice or text input passes through fuzzy matching to correct typos, then routes to the respond() dispatcher covering 31 command categories (app control, browser tabs, file search, music, timers, unit conversion, Wikipedia, and more).

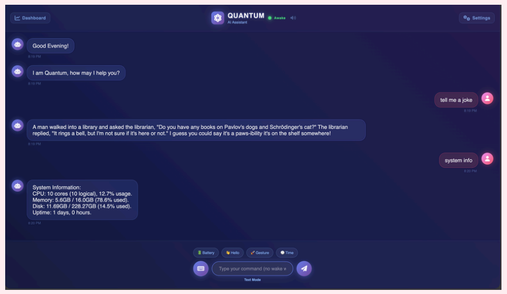

Figure 3: Quantum chat UI with command suggestions

4 LLM Integration & Conversation Memory:
Unrecognized commands are routed to a LangChain agent backed by Gemini 2.5 Flash, Groq LLaMA-3.3-70B, or Ollama. The agent retains the last 8 conversation turns as memory and has access to 9 action tools (web search, file search, weather, calculator, screenshot, etc.).

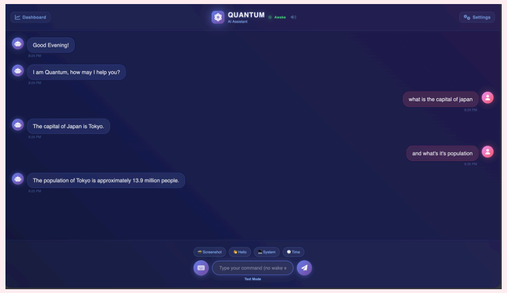

Figure 4:LLM-powered response with conversation context

5. Live Dashboard & Settings Panel:
A browser-based Eel UI displays a real-time dashboard with gesture heatmap, command history, and CPU/RAM stats. The settings panel allows tuning gesture sensitivity, cursor speed, pinch threshold, and scroll speed - changes apply on next gesture session start.

Figure 5 : Live dashboard and settings panel

Conclusion

The Virtual Mouse & Quantum Assistant demonstrates an efficient and innovative approach to human-computer interaction by combining real-time gesture recognition with an AI-powered voice assistant. By leveraging MediaPipe, OpenCV, and LLM integration, the system eliminates dependency on physical input devices while providing intelligent, context-aware command execution. It ensures accessibility, cross-platform compatibility, and extensibility - making it a practical solution for modern computing environments that are shifting toward touchless and voice-driven interaction.

References

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